

Projection

Ek vector ka 'shadow' doosre vector ki direction mein

01 Opening Hook

STORY TIME

Socho dhoop mein ek pencil khadi hai, aur zameen par uska shadow pad raha hai. Shadow pencil ki 'height' ka ek certain fraction hai jo sun ke angle par depend karta hai. Ek vector ka doosre vector par yeh 'shadow' hi hai **Projection**.

02 Why Does This Concept Exist?

THE PROBLEM

Kai baar humein jaanna hota hai ki ek vector doosre vector ki direction mein 'kitna contribute' karta hai – jaise force ka ek component kisi particular direction mein. Projection invent hua taaki hum ek vector ko doosre vector ki direction mein 'measure' kar sakein.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tum ek flashlight seedhe upar se ek stick par chamka rahe ho jo tilt ho ke khadi hai. Zameen par jo shadow padega, uski length stick ki 'projection' hai zameen (ground direction) par.

AI mein bhi, jab hum data ko PCA se kam dimensions mein le jaate hain, hum actually data ko naye axes (eigenvectors) par 'project' kar rahe hote hain.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Projection ek vector ko doosre vector ki direction mein 'flatten' kar deta hai, batakar ki original vector us direction mein kitna 'move' karta hai. Yeh dot product se directly related hai.

05 Ab Thoda Maths

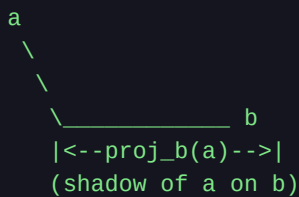
FORMULA TIME

Formula:

$\text{proj}_b(a) = ((a \cdot b) / \|b\|^2) * b$ - yeh vector a ka vector b par projection deta hai.

06 Visualize Everything

PICTURE IT



07 AI Connection

WHERE AI USES THIS

PCA Projection: Data points ko principal components (eigenvectors) par project karke hi PCA dimensionality reduction karta hai.

Attention Mechanism: Query aur Key vectors ke beech 'alignment' dot product/projection se related concept hai jo attention scores banata hai.

08 Real Industry Example

REAL COMPANIES

Google/Meta (Data Science): High-dimensional data ko lower-dimensional space mein project karna (PCA/UMAP) ek standard preprocessing step hai analysis ke liye.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch mein manually: $\text{proj} = (\text{torch.dot}(a,b) / \text{torch.dot}(b,b)) * b$. PCA transform (`pca.transform(X)`) internally data ko principal component vectors par project karta hai.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Projection aur similarity same hain.

Sahi baat: Projection ek vector deta hai (ya uski length), jabki similarity (jaise cosine) ek single normalized number deta hai – dono related hain lekin same nahi.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Vector projection kya hai?

A: Ek vector ka 'shadow' doosre vector ki direction mein – batata hai pehla vector doosre ki direction mein kitna extend karta hai.

[Intermediate] Q: PCA projection ka use kya hai?

A: Data points ko principal component directions par project karke unhe kam dimensions mein represent karna, sabse zyada variance retain karte hue.

[Advanced] Q: Projection aur dot product mein kya mathematical relation hai?

A: Projection formula mein dot product hi numerator hai – dot product batata hai 'kitna alignment' hai, aur usse normalize karke hum exact projection vector nikalte hain.

12 Exercises

PRACTICE TIME

[Easy] Agar a aur b perpendicular hain, unka projection kya hoga?

Solution: Zero vector – kyunki koi overlap nahi hai directions mein.

[Medium] $a=[3,3]$ ko $b=[1,0]$ (x-axis) par project karo.

Solution: $\text{proj} = [3,0]$ (sirf x-component).

[Hard] PCA mein agar hum data ko sirf top-1 principal component par project karte hain, kya trade-off hota hai?

Solution: Hum dimensionality drastically kam kar dete hain (compression), lekin agar top component saara variance capture nahi karta, kuch information loss hota hai.

13 Golden Summary

Concept	Easy Meaning
Projection	Ek vector ka 'shadow' doosre vector ki direction mein
Formula	$((a.b)/ b ^2) * b$
PCA Use	Data ko principal components par project karna
Zero Projection	Perpendicular vectors ka projection zero hota hai
Related Concept	Dot product se directly connected

14 One-Line Memory Trick

“Projection ek vector ka shadow hai doosre vector ki roshni mein.”